CONSTANT SWITCHING FREQUENCY LLC-TYPE SERIES RESONANT CONVERTER

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ABSTRACT

The steady state analysis and performance characteristics of the constant switching frequency, PWM controlled LLC-type SRC are presented for both the continuous and discontinuous conduction mode of operations. It is shown that the control characteristics of the LLC-type SRC are much improved as compared to those of the conventional SRC using the same control technique.

I. INTRODUCTION

There are many techniques available for the output control in resonant converters, which may be broadly classified as variable-frequency [1][2], and constant-frequency controls [3][4]. The variable-frequency control is simple to be implemented in practice, but the EMI noise generated by the converter can be wideband because a large range of switching frequency may be required for output regulation. Other disadvantages associated with the large range of switching frequency are poor cross-regulation in multiple-output converters using multiple-winding transformers, and higher component losses in the upper switching frequencies.

The problems mentioned above can be minimized by using the constant-frequency control technique in the converter design. The constant-frequency control technique can be implemented in two ways; the PWM control and the phase-shift control. In the PWM control, a minimum of two transistor switches are needed, such as those in the half-bridge configuration, and each transistor switch is operated at less than 50% duty ratio. In the phase shift control, the method can be implemented by using two pairs of transistor switches with transistors in each pair being turned off and on at 50% duty ratio. These pairs of switches are arranged so that the phase-shift between them is used for the converter control, such as those in the full-bridge configuration.

It has been shown that, in the case of PWM control, the converter voltage gain of the conventional SRC is load independent only when the converter is operated in the continuous conduction mode [3]. As the output changes from no load to full load in the converter, the converter operation may change from discontinuous to continuous conduction mode. Consequently, the range of duty ratio needed for the output regulation may be prohibitive large. In this paper, we propose a third order converter called LLC-type SRC which will reduce the range of the required duty ratio.

In this paper, we present the analysis of the LLC-type SRC operating in the continuous and discontinuous conduction modes. By using both the time domain and state plane analyses, the steady state expressions can be derived to facilitate the numerical calculation of the control characteristics. To obtain the maximum range of the output voltage gain, the switching frequency is chosen to be the resonant frequency of the converter operating in the continuous conduction mode. We will give the performance characteristics of the converter operating in this condition and compare them with those of the conventional SRC using PWM control.

II. STEADY STATE OPERATION OF THE CONVERTER

An idealized half-bridge LLC-type SRC is shown in Fig. 1. For PWM control, transistors T_1 and T_2 are turned on and off alternately at less than 50% duty ratio. The simplified equivalent circuit is shown in Fig. 2. In this circuit, when the switch S is in either T_1 or T_2 position, diodes D_1 and D_2 are open. When S is in the open position, if either D_1 or D_2 is on, the converter is operated in the continuous conduction mode. Otherwise, it is in the discontinuous conduction mode.

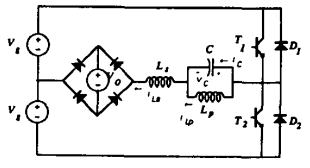

Fig. 1 Half-bridge circuit of the LLC-type series resonant converter.

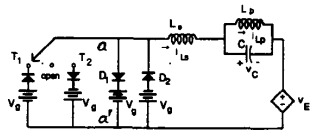

Fig. 2 The converter equivalent circuit.

In the continuous conduction mode of operation, the LLC resonant circuit of the converter is driven by an equivalent two state source $(+V_g, -V_g)$ at terminal a-a' shown in Fig. 2 and is loaded by a two state voltage sink v_E $(+V_o, -V_o)$. Over a switching period, there exist four possible topolgical circuit modes for the converter as indicated in the state-plane diagram given in Fig. 3.

The normalized state-plane diagram for the continuous conduction mode of operation is shown in Fig. 4. The cycle starts at t_0 when T_1 is turned on and the converter is in the circuit mode M1. The circuit mode changes from M1 to M2 at t_1 when transistor T_1 is turned off and i_{Ls} is forced to commutate from T_1 to D_2. In the time interval $[t_1,$

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$t_0 + T_s/2$], both T_1 and T_2 are turned off and switch S in Fig. 2 is in the open circuit position. In circuit mode M2, the amplitude of i_{Ls} decreases and equals to zero at t_2. For the continuous conduction mode of operation, the capacitor voltage v_C should be greater than $V_g + V_o$ at $t = t_2$. Consequently, diode D_1 is naturally turned on at t_2 and the converter is in the circuit mode M4. Diode D_1 stays on until transistor T_2 is turned on at $t = t_0 + T_s/2$. From t_0 to $t_0 + T_s/2$, the switching operation for the one half switching period is completed. The switching events during the other half of switching period can be similarly described.

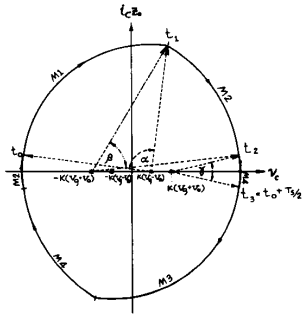


Fig. 3 Steady state trajectory in the continuous conduction mode.

In the discontinuous conduction mode of operation, the equivalent driving source at terminal a-a' has three states $+V_g$, $-V_g$ and open circuit. In this case, the converter possesses two resonant frequencies; ω_o when the driving source equals $+V_g$ or $-V_g$, and ω_p when the driving source is open, where $\omega_o = 1/\sqrt{LC}$, $\omega_p = 1/\sqrt{L_p C}$, and $L = L_p L_s/(L_p + L_s)$. The state plane diagram for the discontinuous conduction mode of operation is shown in Fig. 4.

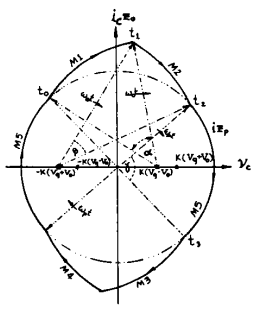


Fig. 4 Steady state trajectory in the discontinuous conduction mode.

The converter operations for the continuous and discontinuous conduction modes are same in the time interval [t_0, t_2]. In the case of discontinuous conduction mode, at $t = t_2$, the capacitor voltage v_C is less than V_g+V_o when i_{Ls} equals zero. Thus, both D_1 and D_2 are turned off and terminal a-a' is open. The residual energy stored in the parallel tank circuit C-L_p will oscillate at resonant frequency ω_p. The converter is in circuit mode M5. At t_3, T_2 is turned on and i_{Ls} increases. Up to this time, the operation phenomena in one half of a switching period is completed. The circuit modes M1, M2, M3, M4, and M5 are shown in Fig. 5.

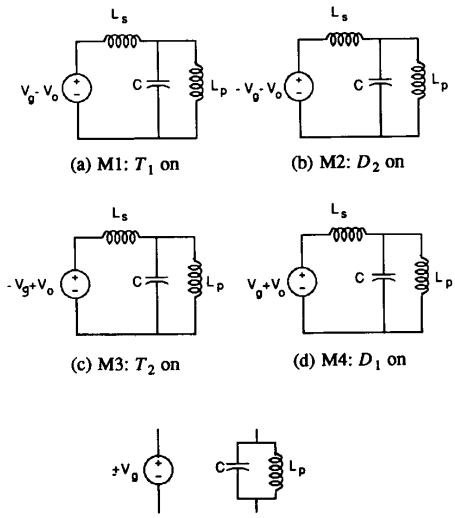


Fig. 5 Five topological circuit modes.

III. STEADY STATE ANALYSIS

In the equivalent circuit shown in Fig. 2, the voltage source has three states; $+V_g$, open circuit, and $-V_g$. The states $+V_g$ and $-V_g$ is controlled by the duty ratio D, otherwise the source is in the open circuit state. When the voltage source is in the open circuit state, the magnitude of inductor current i_{Ls} decreases and either diode D_1 or D_2 conducts, depending on the direction of nonzero current i_{Ls}. When i_{Ls} reaches zero, either both diodes are off or the natural commutation occurs between diodes D_1 and D_2, depending on the value of the capacitor voltage v_C at that time instant. If $|v_C| < V_g + V_o$, both diodes D_1 and D_2 are off and i_{Ls} remains zero until the open circuit state is removed. Otherwise, the natural commutation will take place.

As shown in Fig. 3 and Fig. 4, the mode sequences for the continuous and discontinuous conduction mode of operations are M1 - M2 - M4 - M3 - M4 - M2 and M1 - M2 - M5 - M3 - M4 - M5, respectively. The transition from M2 to M4 or M2 to M5, which determines whether the converter is operated in the continuous or discontinuous conduction mode, depends on whether v_C is greater or less than $V_g + V_o$, respectively, at the instant when i_{Ls} equals zero. In the case of discontinuous conduction mode, there are two resonant frequencies associated with the converter circuit; ω_o in M1, M2, M3, and M4, which is also the natural resonant frequency of the converter, and ω_p in M5. In this paper, our analysis is limited to the case in which the switching frequency is equal to the natural resonant frequency of the converter.

The state equations for circuit modes M1, M2, M3, and M4 are given by,

$$\frac{di_{Ls}}{dt} = \frac{V_s - v_C}{L_s} \tag{1}$$

$$\frac{di_{Lp}}{dt} = \frac{v_C}{L_p} \tag{2}$$

$$\frac{dv_C}{dt} = \frac{i_{Ls} - i_{Lp}}{C} \tag{3}$$

where V_s equals to $V_g - V_o$, $-V_g - V_o$, $-V_g + V_o$, and $V_g + V_o$ for the modes M1, M2, M3, and M4 respectively. From (1), (2), and (3), the state-plane equation for circuit modes M1, M2, M3, and M4 can be written as

$$\frac{di_{nC}}{dv_{nC}} = \frac{KV_{ns} - v_{nC}}{i_{nC}} \tag{4}$$

where $K = \frac{L_p}{L_s + L_p}$, $i_{nC} = \frac{i_C Z_o}{V_g}$, $i_C = i_{Ls} - i_{Lp}$, $v_{nC} = \frac{v_C}{V_g}$,

$$M = \frac{V_o}{V_g}, \quad Z_o = \sqrt{\frac{L_p L_s}{(L_p + L_s)C}},$$

and V_{ns} is the normalized voltage which equals to 1-M, -1-M, -1+M, and 1+M for the mode M1, M2, M3, and M4 respectively. In the discontinuous conduction mode, the state equations for M5 are given by,

$$i_{Ls} = 0 \tag{5}$$

$$\frac{di_{Lp}}{dt} = \frac{v_C}{L_p} \tag{6}$$

$$\frac{dv_C}{dt} = \frac{-i_{Lp}}{C} \tag{7}$$

From (5), (6), and (7), the state plane equation for circuit mode M5 can be written as

$$\frac{di'_{nC}}{dv_{nC}} = \frac{v_{nC}}{-i'_{nC}} \tag{8}$$

where $i'_{nC} = i_C \frac{Z_p}{V_g}$, $Z_p = \sqrt{\frac{L_p}{C}}$.

The steady state trajectory for the discontinuous conduction mode is shown in Fig. 4. In order to determine the transition from M2 to M4 (or from M4 to M2), we need the expression of i_{nLs} ($i_{Ls} Z_o / V_g$) which can be derived from the converter circuit and is given by,

$$\begin{aligned} i_{nLs}(t) = {} & I_{nLs}(t_i)(1 - K + K\cos\omega_o t') + \frac{Z_o}{L_p + L_s} V_{ns}\, t' \\ & + K\frac{Z_o}{\omega_o L_s} V_{ns} \sin\omega_o t' + \frac{Z_o}{\omega_o L_s} V_{nC}(t_i) \sin\omega_o t' \\ & + K\,(I_{nLs}(t_i) - I_{nC}(t_i))\,(1 - \cos\omega_o t') \end{aligned} \tag{9}$$

where $\omega_o = \sqrt{\frac{L_p + L_s}{L_p L_s C}}$, $t' = t - t_i$

and t_i is the instant at which the converter enters into circuit modes M1, M2, M3, or M4. For example, from the state trajectory of the continuous conduction mode shown in Fig. 3, at $t = t_2$, $i_{nLs}(t_2) = 0$. Thus we have,

$$\begin{aligned} i_{nLs}(t_2) = {} & I_{nLs}(t_1)(1 - K + K\cos\beta) \\ & + \frac{Z_o}{L_p + L_s} V_{ns} \frac{\beta}{\omega_o} + \frac{Z_o}{\omega_o L_s} K\, V_{ns} \sin\beta \\ & + K\,(I_{nLs}(t_1) - I_{nC}(t_1))\,(1 - \cos\beta) \\ & + \frac{Z_o}{\omega_o L_s} V_{nC}(t_1) \sin\beta \end{aligned} \tag{10}$$

Equation (10) can be numerically solved to find β which is the conduction angle of diode D_2. The capacitor voltage v_{nC} can also be calculated at $t = t_2$ to determine the next mode sequence M4 or M5 after M2 is terminated. From the trajectory of the continuous conduction mode given in Fig. 3, we have

$$\alpha + \beta + \gamma = \pi \tag{11}$$

From the trajectory of the discontinuous conduction mode shown in Fig. 4, we obtain,

$$\alpha + \beta + \gamma\frac{\omega_o}{\omega_p} = \pi \tag{12}$$

For steady state operation, the following cyclic conditions are satisfied,

$$i_{nLs}(t_0) = -\, i_{nLs}(t_0 + \frac{T_s}{2}) \tag{13}$$

$$v_{nC}(t_0) = -v_{nC}(t_0 + \frac{T_s}{2}) \tag{14}$$

$$i_{nLp}(t_0) = -i_{nLp}(t_0 + \frac{T_s}{2}) \tag{15}$$

Consequently, for a given converter gain M and duty ratio $D = \alpha/\pi$, the steady state solution can be numerically calculated from (13), (14), (15), and other constraints given by (11) and (12). If the solutions of i_{nLs}, v_{nC}, and i_{nLp} exist in the discontinuous conduction mode, they are unique functions of load Q. In the continuous conduction mode, infinite solutions of i_{nLs}, v_{nC}, and i_{nLp} may exist because they are load independent.

IV. CONTROL CHARACTERISTICS

In the series resonant converter, power is coupled from the resonant tank to the output load through the resonant circuit. In the LLC-type SRC, the normalized output current can be obtained from resonant current, i_{nLs}, by using the following relation,

$$I_{no} = \frac{2}{T_s} \int_{t_0}^{t_0 + \frac{T_s}{2}} |i_{nLs}(t)|\, dt \tag{16}$$

The normalized output voltage, which is equal to the converter gain, is equal to,

$$V_{no} = M = \frac{I_{no}}{Q} \tag{17}$$

where Q is the load quality factor. Substituting (17) into (16), we have,

$$Q = \frac{\omega_o}{M\pi} \int_{t_0}^{t_0 + \frac{T_s}{2}} |i_{nLs}(t)|\, dt \tag{18}$$

In terms of current responses in different circuit modes, equation (18) can be written in the forms shown in equations (19) and (20),

$$Q = \frac{\omega_o}{M\pi} \sum_{i=0}^{2} \int_{t_i}^{t_{i+1}} |i_{nLs}(t)|\, dt \tag{19}$$

$$Q = \frac{\omega_o}{M\pi} \sum_{i=0}^{1} \int_{t_i}^{t_{i+1}} |i_{nLs}(t)|\, dt \tag{20}$$

for the continuous and discontinuous conduction modes, respectively. Substituting (9) into (19) and (20), we obtain two expressions which have the following functional form,

$$Q = Q(\,\alpha, \beta, i_{nLs}(t_0), i_{nC}(t_0), v_{nC}(t_0), M\,) \tag{21}$$

From the geometric properties of the state-plane diagrams given in Fig. 3 and 4, it can be shown that following functions can be obtained,

$$\beta = \beta(\alpha, i_{nLs}(t_0), i_{nC}(t_0), v_{nC}(t_0), M) \tag{22}$$

$$\gamma = \gamma(\alpha, \beta) \tag{23}$$

By relating the end points to the starting points for each circuit mode in the state-plane diagrams and from the converter circuits, we obtain three more equations, which satisfy the steady state conditions (13), (14), and (15), in functional forms,

$$i_{nC}(t_0) = f_1(\alpha, \beta, \gamma, i_{nLs}(t_0), i_{nC}(t_0), v_{nC}(t_0), M) \tag{24}$$

$$v_{nC}(t_0) = f_2(\alpha, \beta, \gamma, i_{nLs}(t_0), i_{nC}(t_0), v_{nC}(t_0), M) \tag{25}$$

$$i_{nLs}(t_0) = f_3(\alpha, \beta, \gamma, i_{nLs}(t_0), i_{nC}(t_0), v_{Nc}(t_0), M) \tag{26}$$

In equations (21) to (26), we have eight unknowns in six equations. Specifying any two of unknown parameters, we can solve these equations numerically.

The control characteristics of the PWM controlled LLC-type SRC have been obtained for different values of Q and are given in Fig. 6 for K = 0.5. For a given converter gain and load range, it can be seen from this figure that the control characteristics of the LLC-type SRC are much improved as compared to those of the conventional SRC. For example, for K = 0.25, M = 0.4 and a load range from $6/\pi$ to infinite, the control duty ratio is from 0.375 to 0.5. In the conventional SRC, the range of duty ratio is 0.21 to 0.5. It is about 50% reduction in the control effort to regulate the gain over the same load range.

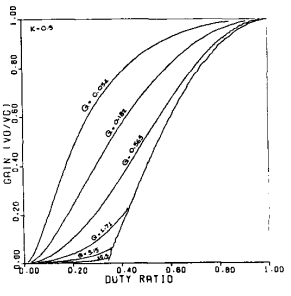
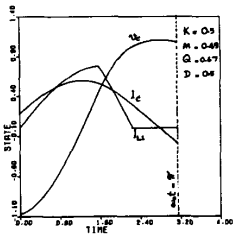


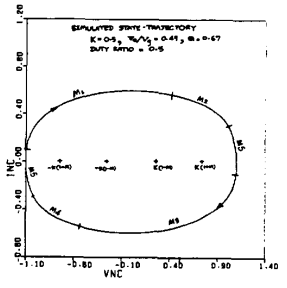
Fig. 6 Control characteristics curves for the LLC-type resonant converter

V. CONCLUSION

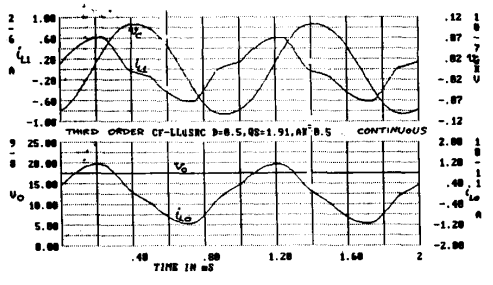
It has been shown that, in the continuous conduction mode of operation, the control characteristics of the LLC-type SRC covers a wider load range for a given voltage gain as compared to the characteristics of the conventional SRC using PWM control. In the discontinuous conduction mode of operation, the control characteristics curves of the LLC-type SRC possesses narrower range of duty ratio as compared to the curves of the conventional SRC for the same load range and voltage gain.



(a) v_C, i_{L1}, and V_o in steady state.



(b) Simulated trajectory of i_{nC} versus v_{nC}.



(c) Simulated response in the continuous conduction mode.

Fig. 7 Simulated waveforms and trajectory.

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